

Can a smooth fluid become infinitely rough?

The 3D Navier–Stokes equations describe water and air. The Millennium Problem asks whether every smooth, physically reasonable starting flow stays smooth for all time—or whether a singularity can form in finite time.

THE QUESTION IN ONE LINE

Smooth forever or **finite blow-up?**

No generally accepted proof decides this in three dimensions.

Our repository investigates one proof route computationally.

It has not solved the Millennium Problem.

Stretching creates small scales; viscosity removes them

The unresolved issue is whether the smoothing always wins before any derivative becomes unbounded.

$$\partial \mathbf{u} / \partial t + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0$$

velocity changes = transport + pressure + viscous smoothing; incompressibility forbids sources or sinks

VORTEX STRETCHING

Can sharpen gradients

- Three-dimensional vortices can stretch and intensify.
- Energy may move toward smaller spatial scales.
- A proof must rule out uncontrolled concentration.

NONLINEAR TERM

$$(\mathbf{u} \cdot \nabla) \mathbf{u}$$

UNKNOWN BALANCE

VISCOSITY

Smooths the flow

- Diffusion damps high-frequency motion.
- Energy decreases in the classical energy estimate.
- Known estimates are not yet strong enough to close 3D regularity.

DIFFUSIVE TERM

$$\nu \Delta \mathbf{u}$$

A successful proof must control every scale uniformly—not only every scale tested by a computer.

A fast falsification laboratory—not an automated proof

The C++ engine converts proof ideas into exact finite tests, searches for hostile states and records enough provenance to restart the analysis.

LAYER	WHAT THE PROJECT DOES	CURRENT RESULT	STATUS
Spectral engine	Finite Fourier / Galerkin Navier–Stokes algebra	Reproducible C++20 kernels and certificates	VERIFIED
Gradient engine	JVP/VJP, RK4 adjoint checks and L-BFGS search	Fast adversarial state optimization	VERIFIED
Far-tail analysis	Separate sufficiently distant frequency shells	Cutoff-independent partial lemma	PROVED
PNT-13	Test standalone shell decorrelation	Near-perfect gap correlations found	FALSIFIED
PNT-12	Keep tail mass, Gram row and normalization coupled	Finite record 4.18215e-4; no theorem	OPEN

WHY THIS CAN SHORTEN RESEARCH

A weak lemma can be attacked in seconds or minutes, its counterexample stored, and the next candidate redesigned around the structure that survived.

WHAT REMAINS

Prove PNT-12, close PNT-4 and RQ-11, then complete the L4 to L5 to L6 chain.